The Frozen Universe Illusion:

Temporal Resolution as a Fundamental Limit on Cosmological Observation and SETI

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Abstract

The observable Universe—pervaded by galactic mergers, stellar evolution, large-scale structure formation, and metric expansion—appears to the human observer as an essentially static tableau. This paper establishes that the perceived cosmic stasis is a *fundamental epistemological constraint* arising from temporal resolution mismatch (TRM) between the characteristic timescales of astrophysical processes and the observational window of the human lifespan. We introduce the Temporal Resolution Ratio ($\mathcal{R} = \tau_{\rm obs}/\tau_{\rm sys}$), derive a perceptibility function from signal-detection theory, and prove a no-go theorem: for any biological observer with finite measurement precision, processes with $\mathcal{R} \ll 1$ are observationally indistinguishable from stasis. The Lifespan-Cosmic Mapping (LCM) maps the 13.8×10^9-year cosmic age onto one human lifetime, yielding a cognitive framework where the galactic year $\approx$ one human year, light traverses 5.47 ly per scaled second, and the distance to Andromeda reduces to 5.4 scaled days. We define a LCM Observability Index and apply it to real astronomical programs (Gaia, LSST, JWST, pulsar timing arrays). The Anthropic Temporal Window (ATW) hypothesis—derived from the Margolus–Levitin bound, metabolic scaling theory, and stellar habitability constraints—predicts that biological observers are confined to ~ 7

of the ~ 61 logarithmic orders of temporal scale, making the Frozen Universe Illusion universal among biochemical civilizations. We construct quantitative toy models demonstrating FUI as a structural observational bias, propose the SETI Temporal Mismatch parameter as a new dimension of the Drake equation, and analyze the Bayesian epistemological implications of reconstructing 1.38×10^{10} years of dynamics from a ~ 400-year observational baseline. Explicit falsifiability conditions and counterargument–response pairs accompany each major claim.

Keywords: temporal resolution, cosmic stasis, observer epistemology, anthropic scaling, Lifespan-Cosmic Mapping, frozen universe illusion, SETI, timescale hierarchy

Contents

1 Introduction

When a human observer raises their eyes to the night sky, they are confronted with what appears to be an immutable arrangement of luminous points—a celestial vault that has remained apparently unchanged since the earliest astronomical records of Mesopotamian civilizations. The constellations cataloged by Hipparchus circa 129 BCE remain recognizable today. To the unaided eye—and to sophisticated modern instrumentation operating within the span of a single human life—the cosmos is, for all practical purposes, *frozen*.

This perception is illusory. The Universe is a theater of extraordinary dynamism. Galaxies collide over timescales of hundreds of millions of years. Stars are born and die in thermonuclear cataclysms. The cosmic web of filaments and voids evolves under dark energy and gravity. Space itself expands (Silk, 2019; Planck Collaboration, 2020). Even the apparently immutable stellar proper motions are now precisely measured by Gaia (Gaia Collaboration, 2023), and time-domain surveys such as ZTF and the forthcoming Rubin Observatory LSST reveal transient phenomena at unprecedented rates (Bellm et al., 2019; Ivezić et al., 2019). Cosmological simulations (IllustrisTNG, EAGLE) reproduce billions of years of galaxy evolution in silico (Pillepich et al., 2018; Conselice, 2014).

Yet all of this knowledge is *inferred*, not directly witnessed. The central question this paper addresses is: **why** does the Universe appear static to the human observer, and what formal framework can account for this epistemological limitation? We argue that the answer lies not in the properties of the cosmos but in the properties of the observer. Specifically, the *temporal resolution* of the human observational apparatus—the ratio of the observer's lifespan to the characteristic timescale of the process under observation—is the governing parameter. When this ratio is vanishingly small, dynamical evolution becomes imperceptible, producing what we term the **Frozen Universe Illusion** (FUI).

The observation that cosmic timescales dwarf human lifespans is, of course, well known and appears in countless popular accounts (Sagan, 1977; Rees, 2000; Carroll, 2010). The philosophy of cosmological observation has been explored from various angles—including the underdetermination of dark energy models (Wolf and Pereira, 2023; Wolf and Duerr, 2024; Duerr and Wolf, 2025), the epistemology of anthropic reasoning (Bostrom, 2002; Azhar and Linnemann, 2025; Helbig, 2023; Friederich, 2017), and the observer reference class problem (Bostrom, 2002). What has been *lacking* is a formal, quantitative framework that: (a) defines the precise conditions under which the illusion holds and breaks; (b) proves impossibility-type results bounding what any biological observer can detect; (c) derives the observer's temporal position from physical first principles; and (d) connects the temporal-resolution constraint to specific methodological consequences for cosmology, SETI, and astrobiology. This paper provides such a framework.

The scaling intuition originates in Kriger (2024a,b), who proposed mapping the age of the Universe onto a human lifespan as a cognitive tool. Independent of this work, Logsdon

(2025) has recently proposed a "Resolution Cosmology" framework examining the role of conscious observation in finalizing physical reality—a complementary perspective that shares our emphasis on the observer's constitutive role in cosmological knowledge. Here we formalize that intuition into a rigorous mathematical apparatus.

1.1 Guiding Principles

The following principles organize the paper's argument:

Principle 1 (Temporal Resolution Principle)**.** Whether a dynamical process appears "static" or "dynamic" is not an intrinsic property of the process but a relational property determined by the ratio $\mathcal{R} = \tau_{\text{obs}}/\tau_{\text{sys}}$ of the observer's lifespan to the system's characteristic timescale.

Principle 2 (Anthropic Temporal Confinement)**.** Any observer based on carbon chemistry, sustained by stellar radiation, and produced by Darwinian evolution on a planetary surface is confined to a narrow band (~ 7 of ~ 61 logarithmic orders) on the hierarchy of physically realizable timescales.

Principle 3 (Temporal Communication Constraint)**.** Two civilizations whose operational timescales differ by more than ~ 6 orders of magnitude cannot engage in real-time communication; their signals are mutually indistinguishable from noise or stasis.

1.2 Scope and Contributions

(i) **Temporal Resolution Ratio and the Frozen Universe Theorem:** A no-go result proving that biological observers cannot detect cosmic dynamics when $\mathcal{R} \ll 1$ (Section 2).

(ii) **Structural Bias Toy Model:** A quantitative Monte Carlo demonstration that FUI emerges as a systematic observational bias, not a psychological quirk (Section 2.6).

(iii) **Lifespan-Cosmic Mapping with Observability Index:** A formal rescaling with proven uniqueness properties, plus a practical index applied to real surveys (Section 3).

(iv) **Anthropic Temporal Window:** Derived from the Margolus–Levitin bound, metabolic scaling, and stellar habitability, with explicit falsification conditions (Section 4).

(v) **SETI Temporal Mismatch:** A new dimension of the Drake equation, with a quantitative detectability model (Section 6.3).

(vi) **Bayesian Epistemology of Snapshot Cosmology:** An analysis of how TRM constrains the posterior distribution over cosmic histories (Section 6.1).

1.3 Structure of the Paper

Section 2 develops the TRM formalism, perceptibility function, and proves the Frozen Universe Theorem and Stroboscopic Theorem. Section 3 constructs the Lifespan-Cosmic Mapping. Section 4 derives the Anthropic Temporal Window. Section 5 examines temporal self-similarity. Section 6 discusses implications for epistemology, SETI, and astrobiology. Section 7 states methodological consequences. Appendices provide computational code.

1.4 Levels of Claim

To avoid conflating mathematical results with empirical assertions, we classify the paper's contributions explicitly:

(a) **Mathematical results** (proven from definitions): The Frozen Universe Theorem (Theorem 2.5), the General Perceptibility Lemma (Lemma 2.4), the Stroboscopic Theorem (Theorem 2.8), the properties of LCM (Propositions 3.2 and 3.3). These are deductive consequences of the definitions and models; their empirical content enters through parameter choices (α, τ_{obs}, τ_{sys}), not through the inequalities themselves.

(b) **Physically motivated conjectures** (supported by order-of-magnitude reasoning and toy models, but not rigorously derived from first principles): The Anthropic Temporal Window hypothesis (Section 4), the temporal self-similarity conjecture (Section 5). These are revisable in light of new evidence; explicit falsification conditions are stated.

(c) **Speculative implications** (logically downstream of the framework, but dependent on unverified assumptions): The SETI temporal mismatch analysis (Section 6.3), the temporal ecology of intelligence (Section 6.4), the temporal Fermi paradox suggestion. These are intended as hypothesis-generating contributions, not established conclusions.

2 Theoretical Framework: Temporal Resolution Mismatch

2.1 Fundamental Definitions

Definition 2.1 (Observer Timescale)**.** Let an observer $\mathcal{O}$ possess a finite contiguous observational interval $\tau_{\text{obs}} \in \mathbb{R}^+$. For a biological human observer, $\tau_{\text{obs}} \lesssim 10^2$ yr. For a sustained observational program, τ_{obs} may extend to $\sim 10^3$–10^4 yr.

Definition 2.2 (System Characteristic Timescale)**.** Let a dynamical process $\mathcal{P}$ have state variable $X(t)$. The characteristic timescale is

$$\tau_{\text{sys}} := \left| \frac{X}{\dot{X}} \right|, \tag{1}$$

the time required for order-unity fractional change. This follows standard usage in dynamical systems theory; it captures the timescale over which external observers could detect significant evolution. For oscillatory systems where $\dot{X}$ passes through zero, $\tau_{\rm sys}$ should be understood as referring to the secular evolution or the envelope of the oscillation, not the instantaneous rate at a turning point.

Definition 2.3 (Temporal Resolution Ratio)**.** The **Temporal Resolution Ratio** (TRR) is

$$\boxed{\mathcal{R} \;:=\; \frac{\tau_{\rm obs}}{\tau_{\rm sys}}} \tag{2}$$

This dimensionless quantity governs whether the observer can resolve the evolution of $\mathcal{P}$.

Equation (2) captures a simple but consequential idea: whether a process appears "dynamic" or "frozen" is a *relational* property of the observer–system pair. A galaxy merger ($\tau_{\rm sys} \sim 10^9$ yr) is frozen for a human ($\mathcal{R} \sim 10^{-8}$) but fully dynamic for a hypothetical observer with a 10^{10}-year lifespan ($\mathcal{R} \sim 10$).

2.2 The Perceptibility Function

Suppose $X(t)$ evolves monotonically, with total change $\Delta X_{\rm max}$ over $\tau_{\rm sys}$. In the observation window $\tau_{\rm obs}$, the observed change is:

$$\Delta X_{\rm obs} \;\approx\; \Delta X_{\rm max} \cdot \mathcal{R} \qquad (\text{for } \mathcal{R} \ll 1). \tag{3}$$

For Gaussian measurement noise with standard deviation σ, the **perceptibility** is the probability that the observed change exceeds the noise:

$$P(\mathcal{R}) \;=\; 1 - \Phi\left(\frac{1}{\alpha\,\mathcal{R}}\right), \tag{4}$$

where Φ is the standard normal CDF and $\alpha := \Delta X_{\rm max}/\sigma$ is the intrinsic signal-to-noise ratio. For a general saturating model:

$$\boxed{P(\mathcal{R}) \;=\; 1 - \exp(-\alpha\,\mathcal{R})} \tag{5}$$

Both forms share the same qualitative behavior: $P \to 0$ as $\mathcal{R} \to 0$ (frozen), and $P \to 1$ as $\mathcal{R} \to \infty$ (dynamic). The exponential form is plotted in Figure 1 for several values of α.

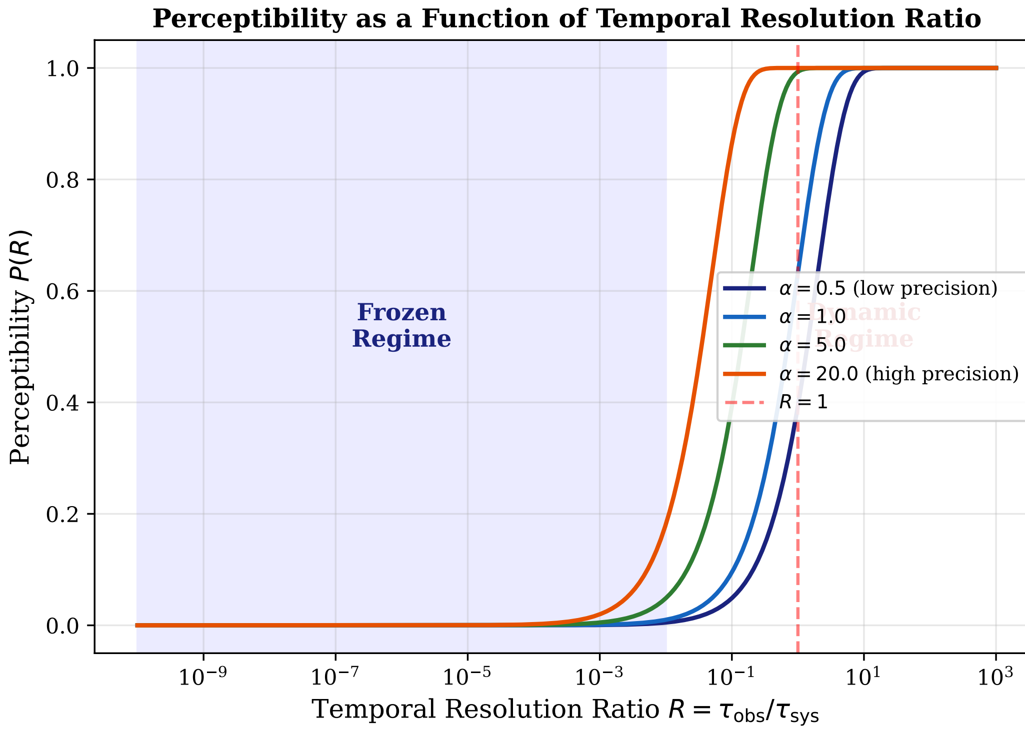

Figure 1: Perceptibility $P(\mathcal{R})$ as a function of the Temporal Resolution Ratio for several values of α. When $\mathcal{R} \ll 1$, perceptibility vanishes regardless of measurement precision.

2.3 The Frozen Universe Theorem (No-Go Result)

Before stating the main result, we establish a model-independent lemma showing that the linear bound on perceptibility holds for *any* reasonable detection model, not just the exponential form in Equation (5).

Lemma 2.4 (General Perceptibility Bound)**.** *Let $P(\mathcal{R})$ be any perceptibility function satisfying: (i) $P(0) = 0$, (ii) P is differentiable at $\mathcal{R} = 0$ with finite derivative $P'(0) = \beta < \infty$, and (iii) $P(\mathcal{R}) \leq 1$ for all $\mathcal{R}$. Then for $\mathcal{R} \ll 1$:*

$$P(\mathcal{R}) \;=\; \beta\,\mathcal{R} + O(\mathcal{R}^2) \;\leq\; \beta\,\mathcal{R} + \varepsilon$$

for arbitrarily small $\varepsilon > 0$, provided $\mathcal{R}$ is sufficiently small.

Proof. By the definition of differentiability, $P(\mathcal{R}) = P(0) + P'(0)\,\mathcal{R} + o(\mathcal{R}) = \beta\,\mathcal{R} + o(\mathcal{R})$. For any $\varepsilon > 0$, there exists $\delta > 0$ such that $|P(\mathcal{R}) - \beta\,\mathcal{R}| < \varepsilon$ for all $\mathcal{R} < \delta$. □

The significance of this lemma is that the Frozen Universe Theorem below does not depend on the specific exponential model: it holds for *any* smooth perceptibility function that starts at zero and has finite initial slope. The exponential form in Equation (5) is a convenient phenomenological choice, but the no-go result is robust to the model.

Theorem 2.5 (Frozen Universe Theorem)**.** *Let $\mathcal{O}$ be an observer with lifespan τ_{obs}, and let $\mathcal{P}$ be a dynamical process with characteristic timescale τ_{sys} and finite intrinsic signal-to-noise ratio $\alpha < \infty$. If $\mathcal{R} = \tau_{\text{obs}}/\tau_{\text{sys}} \ll 1$, then for any perceptibility function satisfying the conditions of Lemma 2.4:*

$$P(\mathcal{R}) \;\leq\; \beta\,\mathcal{R} + \varepsilon \;\ll\; 1,$$

and $\mathcal{P}$ is observationally indistinguishable from a static configuration throughout the observer's entire lifetime. For the exponential model specifically, $\beta = \alpha$ and the bound tightens to $P(\mathcal{R}) \leq \alpha\,\mathcal{R}$.

Proof. Immediate from Lemma 2.4. For the exponential model: $P(\mathcal{R}) = 1 - \exp(-\alpha\,\mathcal{R})$, so $P'(0) = \alpha$. Taylor expansion gives $P = \alpha\,\mathcal{R} - (\alpha\,\mathcal{R})^2/2 + \cdots \leq \alpha\,\mathcal{R}$, with the inequality holding because all higher-order correction terms are negative in the alternating series when $\alpha\,\mathcal{R} < 1$. □

Corollary 2.6 (Population-Level Impossibility)**.** *For observers whose technological history spans less than 10^{-8} of the Hubble time, no single-object cosmological evolution (e.g., an entire stellar life cycle from birth to death) can be directly observed. Only population-level, ensemble-based reconstruction is available.*

Proof. Humanity's telescopic era spans $\sim 4 \times 10^2$ yr, giving $\mathcal{R}_{\text{telescopic}} = 4 \times 10^2/(1.38 \times 10^{10}) \approx 2.9 \times 10^{-8}$. A Sun-like star has $\tau_{\text{sys}} \sim 10^{10}$ yr, so $\mathcal{R} \approx 4 \times 10^{-8}$. By Theorem 2.5, $P \leq \alpha \times 4 \times 10^{-8}$. Even with $\alpha = 10^4$ (extremely high precision), $P \leq 4 \times 10^{-4}$: the star's life cycle is unobservable as a continuous process. □

The corollary highlights a methodological fact well known to astronomers: stellar evolution theory is built from *population studies* (comparing stars at different evolutionary stages in a single snapshot) rather than from watching any individual star age. The TRM framework explains *why* this inferential strategy is necessary: it is a direct consequence of $\mathcal{R} \ll 1$.

2.4 Quantitative Application to Astrophysical Processes

Table 1 presents TRR values for $\tau_{\text{obs}} = 80$ yr.

Table 1: Temporal Resolution Ratios for representative processes ($\tau_{\text{obs}} = 80$ yr).

Process	$\boldsymbol{\tau_{\text{sys}}}$ **(yr)**	$\boldsymbol{\mathcal{R}}$	**Perceptual Status**
Stellar main-sequence (Sun-like)	$\sim 10^{10}$	$\sim 8 \times 10^{-9}$	Completely frozen
Cosmic expansion (Hubble time)	$\sim 1.4 \times 10^{10}$	$\sim 5.7 \times 10^{-9}$	Completely frozen
MW–Andromeda merger	$\sim 4.5 \times 10^{9}$	$\sim 1.8 \times 10^{-8}$	Completely frozen
Galactic rotation (Solar orbit)	$\sim 2.25 \times 10^{8}$	$\sim 3.6 \times 10^{-7}$	Frozen
Continental drift	$\sim 10^{8}$	$\sim 8 \times 10^{-7}$	Frozen
Stellar proper motion (typical)	$\sim 10^5$–10^6	$\sim 10^{-3}$–10^{-4}	Marginally detectable
Mercury orbital precession	$\sim 2.3 \times 10^{5}$	$\sim 3.5 \times 10^{-4}$	Marginally detectable
Supernova light curve	~ 0.5	~ 160	Fully dynamic
Gamma-ray burst	$\sim 3 \times 10^{-8}$	$\sim 2.7 \times 10^{9}$	Instantaneous

The striking asymmetry is visible in Figure 2: the vast majority of cosmic processes lie deep in the $\mathcal{R} \ll 1$ regime. Only transient catastrophic events cross into $\mathcal{R} \geq 1$. This creates a systematic observational bias: the cosmos appears overwhelmingly static, punctuated by rare violent exceptions, when in reality it is continuously dynamic at all scales.

Remark 2.7 (Civilizational Timescale). If we extend τ_{obs} from a single human lifespan (80 yr) to the entire span of recorded astronomical history ($\sim 5 \times 10^3$ yr from Babylonian records, or $\sim 4 \times 10^2$ yr from the telescopic era), several processes shift categories. Stellar proper motions move from "marginally detectable" to "clearly detectable" ($\mathcal{R} \sim 10^{-2}$). However, all processes with $\tau_{\text{sys}} \geq 10^8$ yr—galactic rotation, stellar evolution, cosmic expansion—remain firmly frozen even at civilizational timescales: $\mathcal{R} \leq 5 \times 10^{-5}$. The Frozen Universe Illusion is not merely a limitation of individual human lifespans but of civilizational observational programs.

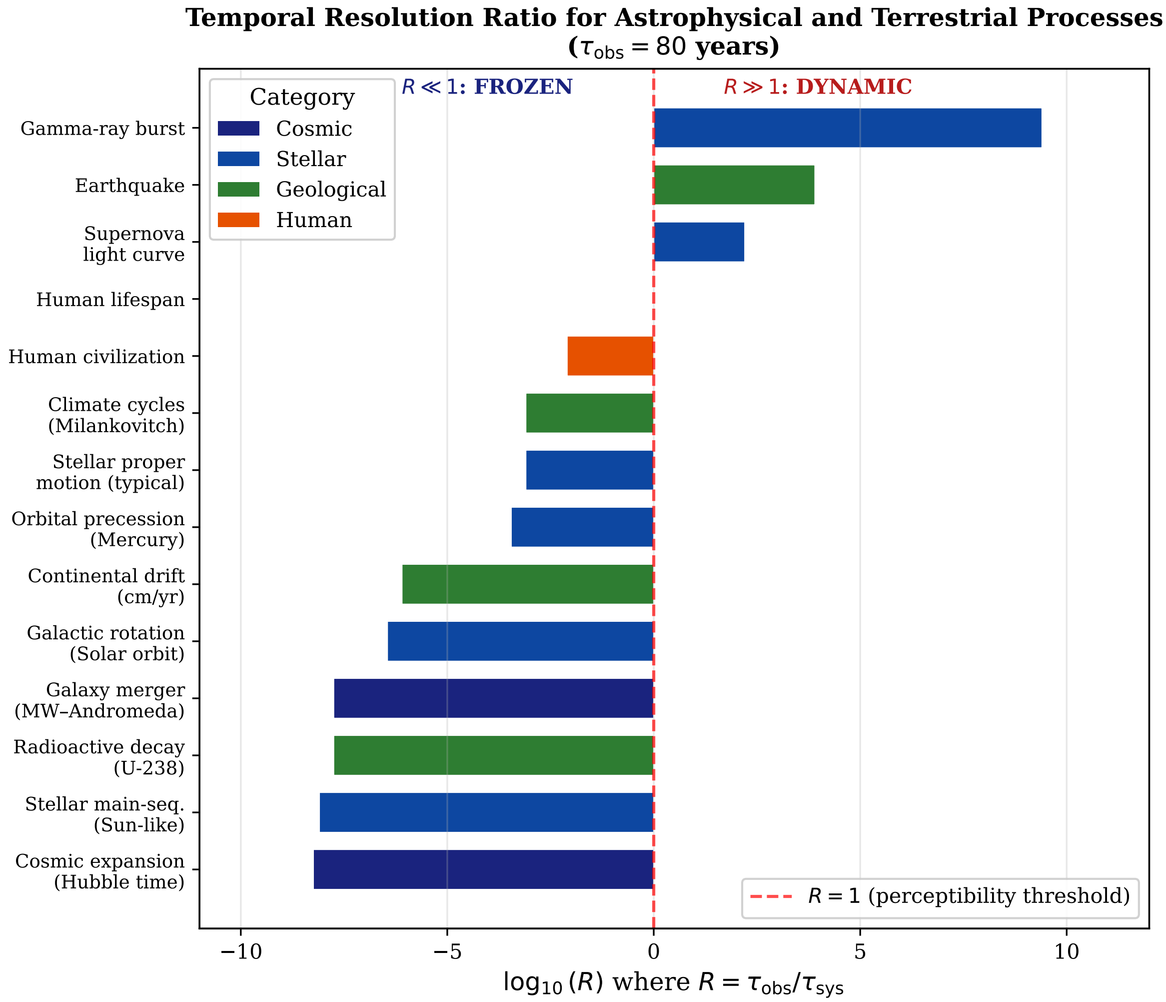

Figure 2: TRR for a range of processes ($\tau_{\mathrm{obs}} = 80$ yr). The dashed line marks $\mathcal{R} = 1$. Nearly all cosmic processes lie in the frozen regime.

2.5 The Stroboscopic Theorem

Theorem 2.8 (Stroboscopic Theorem)**.** *Let $X(t)$ be a band-limited periodic signal with period T, and let the observer sample over $\tau_{\mathrm{obs}} \ll T$. Then the sampled signal is consistent with a constant, and the dynamics are unrecoverable without prior information.*

Proof. By the Nyquist–Shannon theorem (Shannon, 1949; Nyquist, 1928), reconstructing a signal of frequency $f = 1/T$ requires a baseline $\geq T/2$. When $\tau_{\mathrm{obs}} \ll T$, the Taylor expansion of $X(t_0 + \delta t)$ over $\delta t \in [0, \tau_{\mathrm{obs}}]$ gives total variation $|\Delta X_{\mathrm{obs}}| \sim \Delta X_{\max} \cdot (\tau_{\mathrm{obs}}/T) = \Delta X_{\max} \cdot \mathcal{R} \ll \Delta X_{\max}$. The data are consistent with $X(t) = \text{const}$. $\square$

For galactic rotation ($T \approx 225$ Myr), resolving one full cycle requires an observational baseline of ≥ 112 Myr—approximately 1.4×10^6 human lifetimes. Figure 3 illustrates this effect.

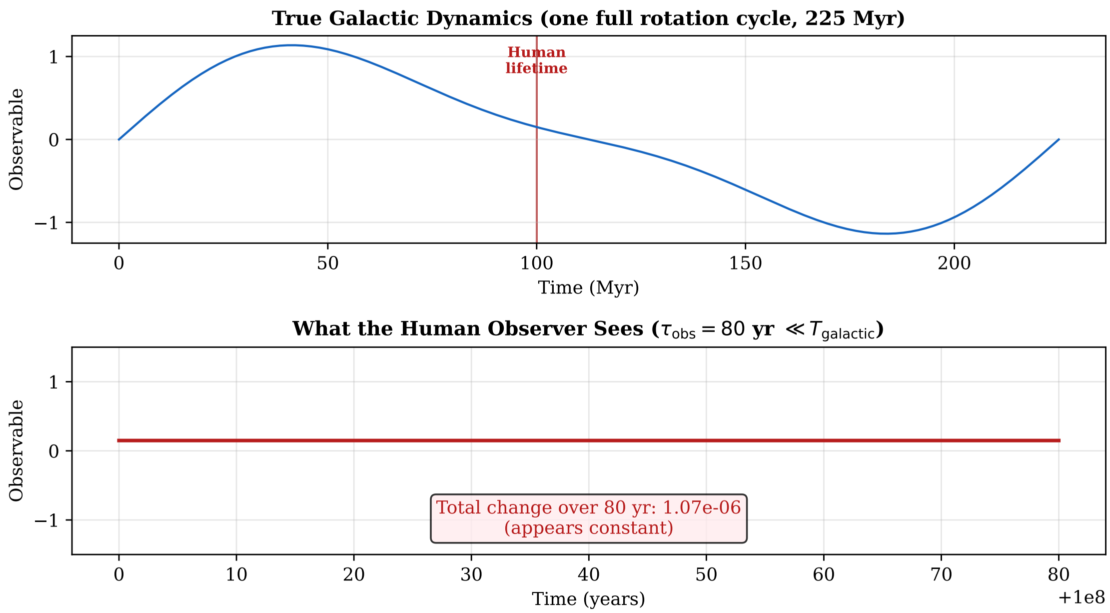

Figure 3: Stroboscopic effect: galactic rotation sampled over 80 yr. Top: full cycle (225 Myr). Bottom: the human-observed signal—essentially a flat line.

2.6 Toy Model: FUI as Structural Observational Bias

To demonstrate that FUI is not merely a subjective impression but a *structural bias* in any finite-$\tau_{\rm obs}$ observation, we construct a Monte Carlo toy model.

Toy Model 2.1 (Structural Bias Model)**.** Consider a universe containing $N = 50{,}000$ dynamical processes with characteristic timescales drawn from a log-uniform distribution over $[10^{-5}, 10^{12}]$ yr. An observer with $\tau_{\rm obs} = 80$ yr and measurement precision $\alpha = 10$ attempts to detect each process. A process is detected if a random draw from $[0, 1]$ falls below $P(\mathcal{R}) = 1 - \exp(-\alpha\,\mathcal{R})$.

The "observed universe" is the subset of detected processes. Compare its timescale distribution to the true distribution.

Figure 4 shows the result. The true distribution is log-uniform, but the observed distribution is *severely truncated*: only processes with $\tau_{\rm sys} \lesssim \tau_{\rm obs}$ are detected. The observer's "universe" consists almost entirely of transient phenomena (supernovae, variable stars, orbital motions), while the dominant dynamical modes (galactic evolution, expansion, stellar aging) are invisible. The Frozen Universe Illusion emerges not from any cognitive failure but from the mathematical structure of finite-window sampling applied to a multi-timescale dynamical system.

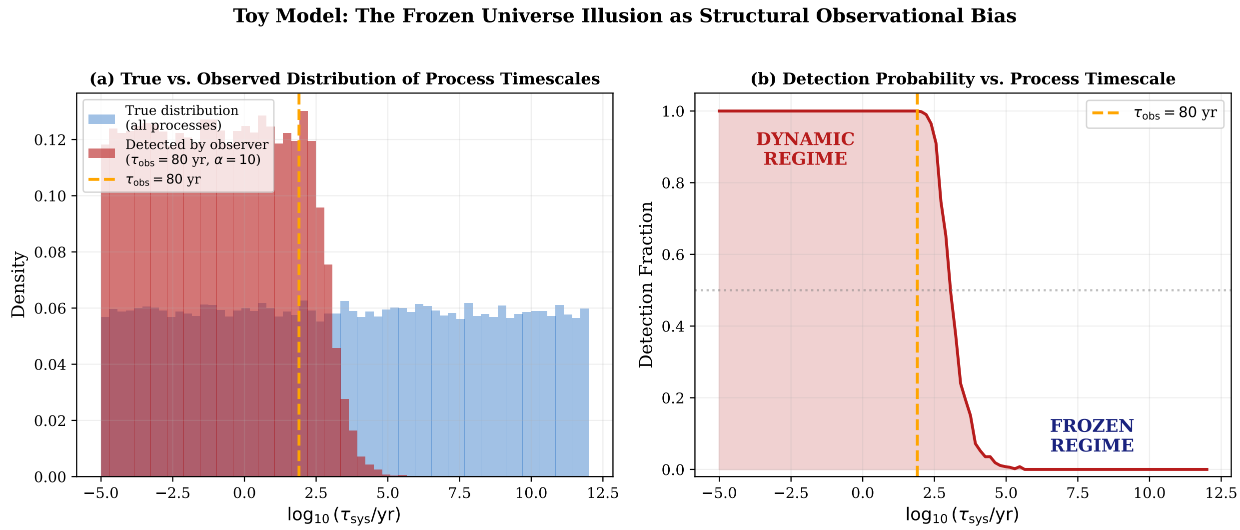

Figure 4: Monte Carlo toy model of the Frozen Universe Illusion as structural observational bias. (a) True vs. observed distributions of process timescales. The observed universe is systematically biased toward short-timescale (transient) phenomena. (b) Detection fraction as a function of process timescale, showing the sharp cutoff at $\tau_{\text{sys}} \sim \tau_{\text{obs}}$.

2.7 Counterarguments and Responses

Counterargument 2.1 ("We observe stellar proper motion—does this not contradict the theorem?").

Response 2.1. Stellar proper motion ($\tau_{\text{sys}} \sim 10^5$ yr) gives $\mathcal{R} \sim 10^{-3}$, placing it in the "marginally detectable" regime. Gaia achieves micro-arcsecond precision ($\alpha \gg 1$), which lowers the effective detection threshold. The Frozen Universe Theorem requires *finite* α; proper-motion detection is fully consistent with the framework and illustrates the boundary between frozen and marginally detectable regimes. The point is that galactic mergers, stellar evolution, and cosmic expansion—the *dominant* dynamical modes—remain firmly frozen regardless of α.

Counterargument 2.2 ("Cosmic expansion is measured via redshift").

Response 2.2. Redshift reveals the *instantaneous rate* of expansion, not its temporal evolution. We observe one snapshot of velocity, not a time-lapse. The deceleration-to-acceleration transition is inferred from supernovae at different redshifts (different *objects* at different epochs), not from watching the same region change speed. This is precisely the "synchronic ensemble inference" the paper distinguishes from "diachronic observation" (Section 6.1).

Counterargument 2.3 ("Gravitational waves reveal dynamical mergers").

Response 2.3. Gravitational-wave events (Abbott et al., 2016) represent the *final seconds* of binary inspirals—precisely the phase where $\mathcal{R} \gg 1$. The preceding 10^6–10^9 yr of orbital decay that brought the binary to coalescence remains unobservable in real time. Gravitational-wave astronomy confirms, rather than contradicts, TRM: we observe only the endpoint where the process timescale finally drops below τ_{obs}.

3 The Lifespan-Cosmic Mapping

3.1 Construction

The Frozen Universe Theorem establishes that cosmic dynamics are imperceptible at human temporal resolution. We now construct a rescaling that restores perceptibility cognitively by mapping cosmic time onto human time.[1]

Definition 3.1 (Lifespan-Cosmic Scaling Factor)**.** Let T_U denote the age of the observable Universe and T_H a representative human lifespan. The scaling factor is

$$\boxed{k \;:=\; \frac{T_H}{T_U} \;=\; \frac{80\ \mathrm{yr}}{13.8\times 10^{9}\ \mathrm{yr}} \;\approx\; 5.80\times 10^{-9}} \tag{6}$$

The model maps any cosmic duration Δt to a scaled duration $\Delta t' = k\cdot\Delta t$, so that $T'_U = k\cdot T_U = T_H = 80$ yr. We extend the scaling to spatial quantities by defining a **scaled speed of light**:

$$c' = \frac{c}{k}. \tag{7}$$

To see what this means concretely: since light travels 1 ly in 1 yr of physical time, it traverses $1/k \approx 1.724\times 10^{8}$ ly in one scaled year. Per scaled second:

$$c' \;=\; \frac{1.724\times 10^{8}\ \mathrm{ly/yr}}{3.156\times 10^{7}\ \mathrm{s/yr}} \;\approx\; 5.47\ \text{ly per scaled second.} \tag{8}$$

The nearest star (Alpha Centauri, 4.37 ly) is less than one scaled second away—cognitively, less than a heartbeat.

3.2 Formal Properties

Proposition 3.2 (Linearity and Order-Preservation)**.** *The map $\mathcal{K} : t \mapsto k\,t$ preserves temporal ordering, duration ratios, and simultaneity.*

Proof. Immediate from $k > 0$ and linearity. □

Proposition 3.3 (TRR Invariance)**.** *The TRR is invariant under LCM: $\mathcal{R}' = (k\,\tau_{\mathrm{obs}})/(k\,\tau_{\mathrm{sys}}) = \mathcal{R}$.*

[1] The scaling intuition originates in Kriger (2024a,b); it is formalized here.

This deserves emphasis: LCM does not change any physical fact or alter the observer's ability to detect cosmic dynamics. It provides a *cognitive coordinate system*—analogous to comoving coordinates revealing homogeneity that is obscured in other gauges—in which the magnitudes of cosmic timescales become intuitively graspable.

Remark 3.4 (Scaling Uniqueness). LCM is the unique linear temporal rescaling satisfying: (i) $T'_U = T_H$, (ii) preservation of duration ratios, and (iii) consistent spatial rescaling via light-travel time. The proof is immediate: condition (i) fixes $k = T_H/T_U$ uniquely; linearity gives (ii); condition (iii) requires $c' = c/k$, uniquely determined by k. This uniqueness is trivial—it follows directly from the defining conditions—but it establishes that LCM is not one of many possible rescalings but the only one satisfying these natural desiderata.

3.3 Rescaled Cosmic Parameters

Table 2: Cosmic quantities under the Lifespan-Cosmic Mapping ($k \approx 5.80 \times 10^{-9}$).

Quantity	**Physical Value**	**LCM-Scaled Value**
Age of Universe	13.8×10^9 yr	80 yr (one human life)
Galactic year (Solar orbit)	225–250 Myr	$\approx$ 1.3–1.45 yr
Sun's main-sequence lifetime	$\sim 10^{10}$ yr	$\approx$ 58 yr
MW–Andromeda merger (from now)	$\sim 4.5 \times 10^9$ yr	$\approx$ 26 yr
Earth's age	4.54×10^9 yr	$\approx$ 26.3 yr
Cambrian explosion	$\sim$ 538 Myr ago	$\approx$ 3.1 yr ago
Dinosaur extinction	$\sim$ 66 Myr ago	$\approx$ 4.6 months ago
Distance to Alpha Centauri	4.37 ly	0.80 scaled seconds
Distance to Andromeda	2.537×10^6 ly	$\approx$ 5.4 scaled days
Observable Universe diameter	$\sim 93 \times 10^9$ ly	$\approx$ 197 scaled days
1 scaled second	5.47 yr	5.47 ly traversed

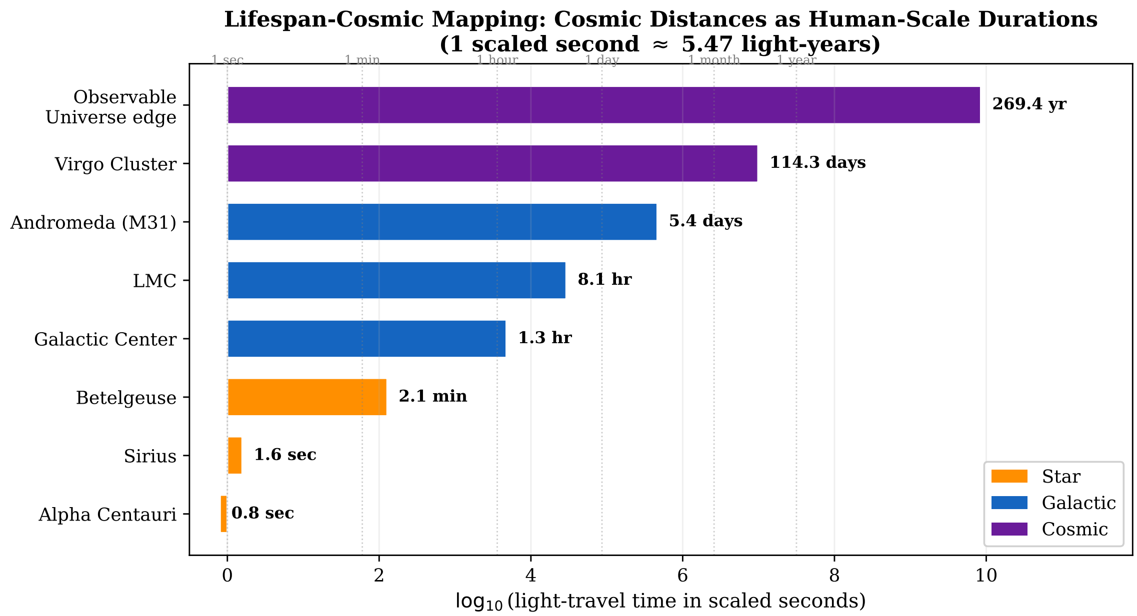

Figure 5: Cosmic distances expressed as human-scale durations under LCM.

3.4 The LCM Observability Index

To give LCM operational content beyond pedagogy, we define a practical index for evaluating astronomical programs.

Definition 3.5 (Observability Index)**.** For an astronomical program with observational baseline τ_b targeting a process with timescale τ_{sys}, the **LCM Observability Index** is

$$\boxed{\mathcal{O} := \frac{\tau_b}{\tau_{\text{sys}}}} \tag{9}$$

Programs with $\mathcal{O} \geq 1$ can directly observe the targeted dynamics; programs with $\mathcal{O} \ll 1$ must rely on ensemble or model-based reconstruction.

Table 3 and Figure 6 apply the index to major astronomical programs.

Table 3: LCM Observability Index for astronomical programs.

Program	τ_b (yr)	**Target** τ_{sys} (yr)	$\mathcal{O}$	$\log_{10} \mathcal{O}$
Gaia DR3 (proper motions)	5.5	10^5	5.5×10^{-5}	-4.3
Hipparcos→Gaia baseline	25	10^5	2.5×10^{-4}	-3.6
ZTF / LSST (transients)	5	10^{-2}	500	2.7
Supernova cosmology (∼40 yr)	40	0.5	80	1.9
Pulsar timing arrays	20	10^8	2×10^{-7}	-6.7
CMB (single epoch)	0.001	1.38×10^{10}	7×10^{-14}	-13.2
Galaxy surveys (30 yr)	30	10^9	3×10^{-8}	-7.5
Entire telescopic era	415	10^{10}	4×10^{-8}	-7.4

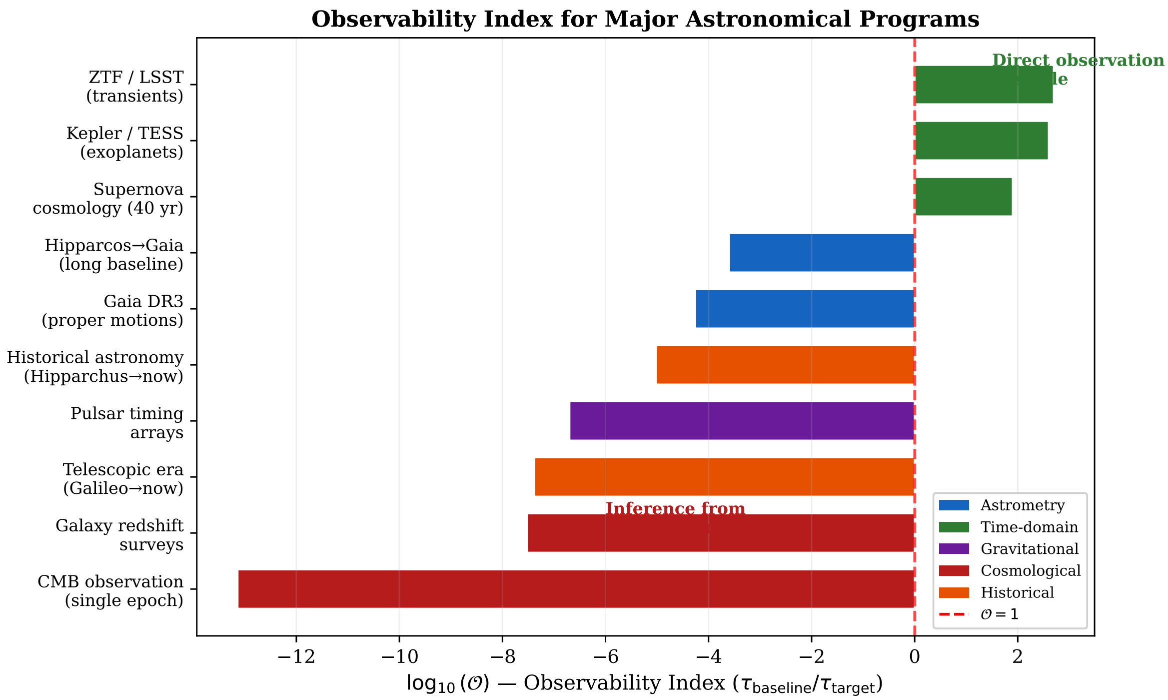

Figure 6: Observability Index for major astronomical programs. Programs left of $\mathcal{O} = 1$ (red line) cannot directly observe their target dynamics and must rely on inference from snapshots.

The Observability Index reveals a methodological hierarchy: time-domain surveys (ZTF, LSST, TESS) operate comfortably in the $\mathcal{O} \gg 1$ regime for their targets; astrometric programs like Gaia achieve $\mathcal{O} \sim 10^{-4}$ for stellar kinematics (marginally detectable with high-precision instruments); but cosmological programs—CMB, galaxy surveys, even the entire 415-year telescopic era—operate at $\mathcal{O} \sim 10^{-8}$ to 10^{-13} for their fundamental targets.

Remark 3.6 (Design Principle). Programs that maximize $\mathcal{O}$ under fixed human time are epistemically privileged: they can distinguish direct observation from model-dependent inference.

This suggests a formal criterion for evaluating the epistemological status of cosmological claims.

Remark 3.7 (Methodological Proposal: Epistemological Grading). We propose that observational claims in cosmology be explicitly graded by their Observability Index:

- **Grade A** ($\mathcal{O} \geq 1$): Direct diachronic observation. The process has been watched unfolding. Examples: supernova light curves, exoplanet transits, variable-star periods.
- **Grade B** ($10^{-4} \leq \mathcal{O} < 1$): Marginally resolved. Change detected, but trajectory not fully sampled. Examples: stellar proper motions (Gaia), binary orbital decay (Hulse–Taylor pulsar).
- **Grade C** ($\mathcal{O} < 10^{-4}$): Synchronic inference only. Dynamics reconstructed from population snapshots plus theoretical models. Examples: stellar evolution, galaxy morphological evolution, cosmic expansion history.

This grading does not judge reliability—Grade C includes some of the most precisely confirmed predictions in physics (ΛCDM concordance). It classifies *epistemological mode*: what kind of evidence supports the claim. Making this explicit would clarify the inferential structure of cosmological arguments and identify where model dependence is irreducible.

3.5 Distinction from Sagan's Cosmic Calendar

The LCM should be compared with the "Cosmic Calendar" of Sagan (1977), which compresses T_U into one calendar year. The models share a pedagogical purpose but differ structurally:

(i) **Anchor:** Sagan maps onto 365 days (a calendrical abstraction); LCM maps onto an entire life (experiential). The cognitive engagement is qualitatively different.

(ii) **Spatial extension:** Sagan rescales only time. LCM rescales space through $c' = c/k$, enabling intergalactic "distances" in human units. This is a straightforward algebraic consequence of the temporal mapping but provides additional cognitive leverage absent from Sagan's model.

(iii) **Purpose:** The Cosmic Calendar dramatizes the *recency of humanity*. LCM is designed to *expose the mechanism of FUI*—showing quantitatively why the cosmos must appear static and what it would "look like" with the illusion removed.

We acknowledge that these are differences of emphasis rather than of fundamental kind. The core rescaling idea—compressing cosmic time into human time—is shared, and Sagan's priority in this concept is not disputed. LCM's contribution is the formal apparatus (uniqueness theorem, observability index, TRR invariance proof) and the specific epistemological application.

3.6 Counterarguments and Responses

Counterargument 3.1 ("LCM is just a unit conversion with no scientific content").

Response 3.1. While LCM preserves all physical ratios (Proposition 3.3), it is more than a unit conversion: it is a *cognitive coordinate system* that exposes structure otherwise hidden by the human temporal gauge. Moreover, the Observability Index (Definition 3.5) gives LCM operational content: it provides a quantitative criterion for classifying the epistemological status of observational programs. This moves LCM from pure pedagogy to a proposed methodological tool.

Counterargument 3.2 ("The galactic-year $\approx$ human-year coincidence is an artifact, not a discovery").

Response 3.2. We agree that this coincidence is an artifact of the specific value of T_U/T_H and carries no physical meaning. We present it as a *mnemonic*, not a result. Its value is purely cognitive: it provides an intuitive anchor for a process ($\sim$ 225 Myr orbital period) that is otherwise entirely abstract.

4 The Anthropic Temporal Window Hypothesis

4.1 Motivation

The Frozen Universe Theorem raises a deeper question: is it accidental that biological observers occupy a temporal scale where $\mathcal{R} \ll 1$ for cosmic processes, or is this a necessary consequence of the physics of complexity? We argue for the latter, deriving the observer's temporal confinement from three independent physical constraints.

4.2 The Hierarchy of Timescales

Physically meaningful timescales span $\sim$ 61 orders of magnitude:

$$\text{Planck time: } t_P = \sqrt{\frac{\hbar G}{c^5}} \approx 5.39 \times 10^{-44}\ \text{s}, \qquad \log_{10}(t_P/\text{s}) \approx -43.3, \tag{10}$$

$$\text{Hubble time: } t_H = H_0^{-1} \approx 4.35 \times 10^{17}\ \text{s}, \qquad \log_{10}(t_H/\text{s}) \approx 17.6. \tag{11}$$

Total range: $\Delta \log \approx 60.9$ orders of magnitude.

4.3 Derivation of ATW Bounds from Physical Principles

4.3.1 Lower Bound: Information-Processing Speed

The Margolus–Levitin theorem (Margolus and Levitin, 1998) establishes that the minimum time for a quantum system with energy E to transition between orthogonal states is

$$\delta t_{\min} = \frac{\pi\hbar}{2E}. \tag{12}$$

A self-replicating information-processing system must execute a minimum number of state transitions to encode, read, and copy its genome. Following Lloyd (2000), the minimum number of operations for a system with n bits of genomic information is $\Omega(n)$. For a minimal genome ($n \sim 10^5$–10^6 bits for free-living organisms; *cf.* Gillooly et al. 2001), with each operation requiring $\delta t_{\min}$, the minimum replication time is:

$$\tau_{\text{info}} \gtrsim n \cdot \delta t_{\min} \sim 10^5 \cdot \frac{\pi\hbar}{2k_B T}, \tag{13}$$

where T is the operating temperature. At $T \sim 300$ K (biochemical conditions):

$$\delta t_{\min} \sim \frac{10^{-34}}{10^{-21}} \sim 10^{-13}\ \text{s}, \qquad \tau_{\text{info}} \sim 10^5 \times 10^{-13} \sim 10^{-8}\ \text{s}. \tag{14}$$

This is the *absolute floor* for a single replication event. A complex organism capable of environmental modeling and theory construction requires many orders of magnitude more operations. The metabolic scaling theory of West et al. (1997); Brown et al. (2004) establishes that organismal lifespans scale as $\tau_{\text{life}} \propto M^{1/4}$, where M is body mass (Kleiber, 1932). For the minimum body mass capable of supporting a nervous system ($M \sim 10^{-6}$ kg, comparable to small insects):

$$\tau_{\min} \sim 10^6\ \text{s} \approx 12\ \text{days}. \tag{15}$$

4.3.2 Upper Bound: Energy Availability and Stellar Habitability

A biological observer requires sustained thermodynamic gradients. For surface-dwelling life, these are provided by the host star. The main-sequence lifetime of a star scales as (Cox, 2000):

$$\tau_\star \propto M_\star^{-2.5}, \tag{16}$$

with $\tau_\star \sim 10^{10}$ yr for a solar-mass star. The *habitable* window—during which conditions permit complex multicellular life—is shorter. Lineweaver (2001) and Kipping (2020) estimate that complex life on Earth arose after $\sim 4 \times 10^9$ yr of stellar evolution, leaving ~ 1–2×10^9 yr of remaining habitability.

Even with generous assumptions about alternative metabolisms, the maximum biologically sustained observer lifespan is bounded:

$$\tau_{\max} \lesssim 10^{13}\ \text{s} \approx 3 \times 10^{5}\ \text{yr}. \tag{17}$$

This bound accommodates hypothetical organisms with extremely slow metabolisms ($M^{1/4}$ scaling extrapolated to the largest conceivable body plans), but cannot extend beyond the stellar habitable window.

4.4 Formal Statement

Definition 4.1 (Anthropic Temporal Window)**.**

$$\boxed{\mathcal{W} := \left\{ \tau \,\middle|\, 10^{6}\ \text{s} \lesssim \tau \lesssim 10^{13}\ \text{s} \right\}} \tag{18}$$

spanning ~ 7 orders of magnitude, or $\sim 11.5\%$ of the full 61-order hierarchy.

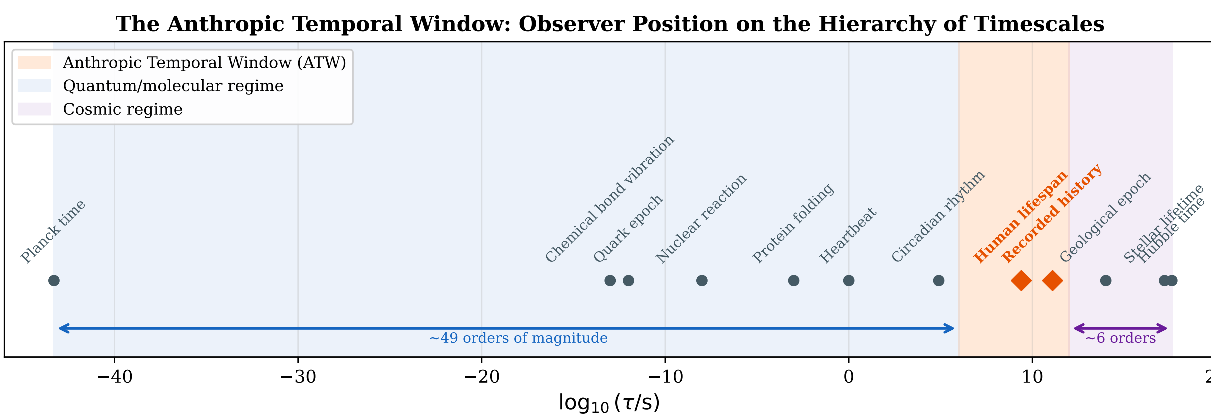

Figure 7: The ATW on the logarithmic hierarchy from Planck to Hubble time.

4.5 Temporal Habitability Function

Figure 8 shows the ATW derived as the product of three constraints: structural stability $S(\tau) = 1 - \exp(-\tau/\tau_{\text{chem}})$, energy availability $E(\tau) = \exp(-\tau/\tau_{\star})$, and information capacity $I(\tau) = 1 - \exp(-\tau/\tau_{\text{info}})$.

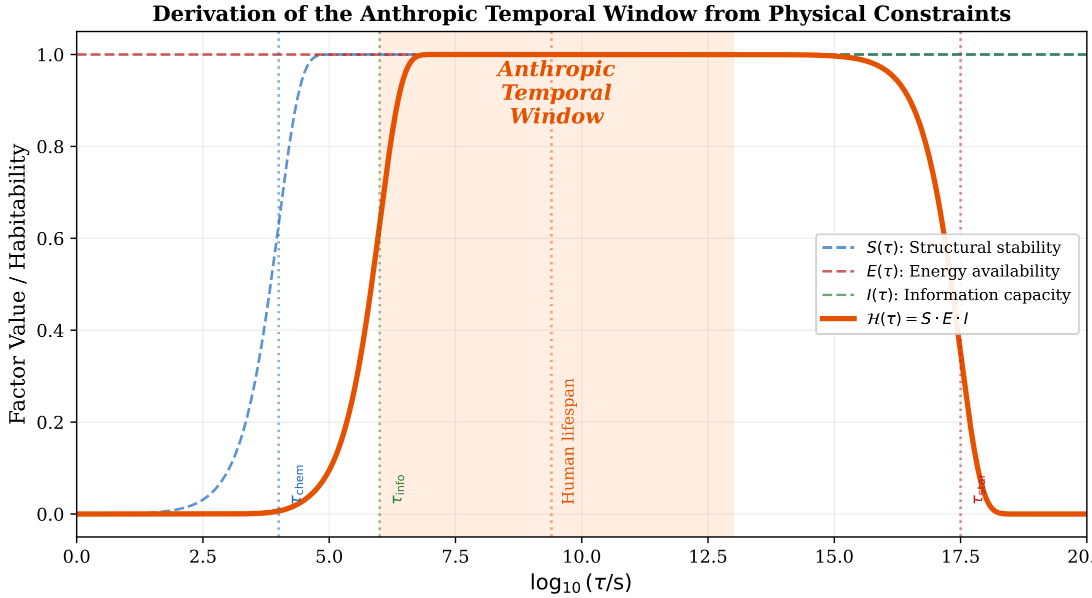

Figure 8: The temporal habitability function $\mathcal{H}(\tau) = S \cdot E \cdot I$, peaking within the ATW. Each constraint is derived from independent physics.

4.6 Universality of the Frozen Universe Illusion

Theorem 4.2 (Universality)**.** *If all biological observers are confined to $\mathcal{W}$, then the Frozen Universe Illusion is universal: every biochemical civilization perceives the cosmos as approximately static.*

Proof. For any $\tau_{\text{obs}} \in \mathcal{W}$, $\tau_{\text{obs}} \leq 10^{13}$ s $\approx 3 \times 10^5$ yr. For cosmic processes with $\tau_{\text{sys}} \geq 10^7$ yr: $\mathcal{R} \leq 3 \times 10^{-2}$. By Theorem 2.5, $P \leq \alpha \times 3 \times 10^{-2}$. For processes with $\tau_{\text{sys}} \geq 10^8$ yr, $P \leq 3\alpha \times 10^{-3}$—negligible for any reasonable α. $\square$

4.7 Relation to Standard Anthropic Arguments

The ATW differs from the anthropic reasoning of Weinberg (1987), Barrow and Tipler (1986), and Bostrom (2002) along a specific axis. Recent philosophical work has reexamined the anthropic principle in light of modern cosmology: Azhar and Linnemann (2025) argue for a reformulation that avoids anthropocentric bias, Helbig (2023) critically examines misconceptions about fine-tuning, and Friederich (2017) addresses the observer reference class problem. Standard anthropic arguments concern the *fine-tuning of physical constants*: why does Λ have its observed value? Why are there three spatial dimensions? Why is $\alpha_{\text{em}} \approx 1/137$? These arguments operate in *parameter space*—the space of possible physical constants.

The ATW operates in *temporal-position space*: given the actual values of physical constants, at what temporal scale must observers exist? This is an orthogonal axis. The ATW does not ask "why do the constants have these values?" but rather "given these constants, what temporal window is available for complexity?" The answer—~ 7 of ~ 61 logarithmic orders—is a derived consequence of known physics (molecular bonding timescales, metabolic scaling laws, stellar evolution), not a free parameter requiring anthropic explanation.

Concretely: Weinberg (1987) showed that Λ must lie within a narrow range for galaxies to form. The ATW shows that, independently of Λ, the observer's *temporal resolution* must lie within a narrow range for biological complexity to persist. These are complementary constraints: Weinberg's constrains the universe's parameters; the ATW constrains the observer's position within the universe's temporal hierarchy.

4.8 Falsifiability and Empirical Status

The ATW is a physically motivated conjecture (Level (b) in Section 1.4), not a theorem. Its empirical content lies in the specific numerical bounds, which are derived from independent physics but are uncertain by ~ 1–2 orders of magnitude. The following observations would falsify the ATW as stated:

(a) **Discovery of fast biological complexity:** A self-reproducing, information-processing system operating stably at $\tau \ll 10^6$ s—for instance, a chemical oscillator network with sub-second replication, heritable variation, and adaptive behavior—would falsify the lower bound.

(b) **Discovery of slow biological observers:** A biological system with characteristic lifespan $\tau \gg 10^{13}$ s (e.g., a geological metabolism sustained over $\sim 10^7$ yr with heritable information and environmental modeling) would falsify the upper bound.

(c) **Demonstration that complexity requires no temporal window:** A rigorous physical argument showing that self-replicating information processors can operate at *any* timescale (including $< 10^{-10}$ s or $> 10^{15}$ s) without requiring structural stability or sustained energy gradients would undermine the ATW's physical basis.

We note what would *not* falsify the ATW: the discovery of post-biological intelligence (AI, plasma vortices) operating outside $\mathcal{W}$. The ATW explicitly applies to *biochemical* observers produced by Darwinian evolution on planetary surfaces. Post-biological counterexamples would confirm the biological specificity of the ATW, not refute it.

4.9 Temporal Reciprocity and Its Asymmetry

Proposition 4.3 (Temporal Reciprocity)**.** *If observer $\mathcal{O}_1$ at timescale τ_1 perceives system $\mathcal{S}$ at timescale $\tau_2 \gg \tau_1$ as frozen, then a hypothetical observer $\mathcal{O}_2$ at timescale τ_2 perceives systems at timescale τ_1 as ephemeral—equally inaccessible, but for a different reason.*

An important distinction must be drawn. The two directions of temporal mismatch produce qualitatively different perceptual failures:

- **Slow-on-fast (FUI):** A slow process observed by a fast observer appears *static*—unchanging, frozen, as though nothing is happening. This is the Frozen Universe Illusion proper.
- **Fast-on-slow:** A fast process observed by a slow observer appears *instantaneous*—a flash, a spike, an event without internal structure. The slow observer does not perceive stasis but rather *ephemerality*: the process is over before it can be examined.

Both directions render the process inaccessible to detailed observation, but the phenomenology differs. A galaxy-timescale observer would not perceive human civilization as "frozen"—it would perceive it as a flash too brief to analyze, if it noticed it at all. Conversely, humans do not perceive quantum processes as ephemeral flashes; most quantum processes are entirely below the threshold of perception, registered only through their aggregate statistical effects on macroscopic observables.

The symmetry lies in the *inaccessibility*: both $\mathcal{R} \ll 1$ and $\mathcal{R} \gg 1$ (from the other observer's perspective) render detailed dynamical observation impossible. The asymmetry lies in the *character* of the failure: stasis versus ephemerality.

4.10 Counterarguments and Responses

Counterargument 4.1 ("Post-biological intelligence could escape the ATW").

Response 4.1. Correct. The ATW applies to *biological* observers produced by Darwinian evolution. Post-biological intelligences could operate outside $\mathcal{W}$; for such entities, FUI might be absent. This delimits ATW's scope but does not invalidate it—it applies to the only class of observer currently known to exist.

Counterargument 4.2 ("ATW is tautological: observers exist where they can exist").

Response 4.2. The charge of tautology conflates two claims. A tautology would be: "observers exist where observers can exist" (trivially true). The ATW asserts something substantive: that the physical constraints of biochemistry, thermodynamics, and stellar astrophysics *jointly confine* observers to ~ 7 of ~ 61 available orders of magnitude, and that this confinement has a specific, testable consequence (universal FUI). The bounds are derived from independent physics (Margolus–Levitin, metabolic scaling, stellar evolution), not from the observation that observers exist. The falsifiability conditions in Section 4.8 further distinguish ATW from tautology.

5 Temporal Self-Similarity: A Conjecture

5.1 Status and Scope

We present temporal self-similarity as an explicitly *conjectural* hypothesis, supported by qualitative parallels and a toy model but not yet by rigorous empirical verification. The section is included because, if confirmed, this conjecture would provide a unifying structural principle connecting the TRM framework across all temporal scales. We are transparent about its speculative character.

5.2 Observational Parallels

Table 4 compares dynamical phases across four temporal regimes.

Table 4: Structural parallels in dynamical phases across temporal scales.

Phase	**Quantum** ($\sim 10^{-15}$ s)	**Biological** ($\sim 10^{9}$ s)	**Geological** ($\sim 10^{15}$ s)	**Cosmic** ($\sim 10^{17}$ s)
Aggregation	Hadron formation	Embryogenesis	Planetary accretion	Protogalactic collapse
Complexification	Nuclear binding	Maturation	Orogeny, biosphere	Large-scale structure
Disruption	Radioactive decay	Senescence, death	Mass extinction	Supernova, merger
Renewal	Daughter nuclei	Reproduction	Adaptive radiation	Star formation

These parallels are qualitative, and we emphasize that the physical mechanisms differ entirely at each scale: the strong nuclear force governs quantum aggregation, gravity governs cosmic aggregation. The claim is not that identical mechanisms operate at all scales, but that a common *structural pattern*—accumulation, organization, catastrophic release, reorganization—manifests repeatedly.

5.3 Toward Quantification: Power-Law Event Spacing

To move beyond heuristic parallels, we identify a quantitative candidate for testing. Consider the distribution of "disruption events" (defined operationally as processes that reset the dominant state variable) across temporal scales. If temporal self-similarity holds, the number density of disruption events per logarithmic time interval should be approximately constant:

$$\frac{dN}{d\log\tau} \approx \text{const}, \tag{19}$$

which implies a power-law distribution of event timescales:

$$p(\tau) \propto \tau^{-1}. \tag{20}$$

This "$1/\tau$" distribution would mean that disruption events are equally likely at every logarithmic scale—a hallmark of temporal self-similarity. Intriguingly, $1/f$ noise (equivalently, $1/\tau$ scaling in time domain) is observed across an extraordinary range of physical systems, from electronic devices to heartbeat variability to river discharge (Mandelbrot, 1982).

Remark 5.1 (Open Problem). Establish whether the frequency of phase-transition events (particle formation, speciation, mass extinction, supernova rate, galaxy merger rate) follows a $1/\tau$ distribution across the full range of temporal scales, or whether characteristic breaks appear at specific scales.

5.4 Toy Model: Mechanism Demonstration

The following toy model does not constitute evidence for temporal self-similarity; it demonstrates the *mechanism* by which self-similarity would arise if a specific physical condition holds. The question of whether that condition actually holds is empirical and unresolved.

Toy Model 5.1 (Self-Similar Temporal Process)**.** Consider a family of systems indexed by τ, each governed by

$$\frac{dX}{dt} = r\,X(1 - X/K) - \frac{\delta\,X}{1 + (t/\tau)^2}, \tag{21}$$

with logistic growth plus Lorentzian disruption. Under rescaling $s = t/\tau$:

$$\frac{dX}{ds} = r\tau\,X(1 - X/K) - \frac{\delta\tau\,X}{1 + s^2}. \tag{22}$$

If $r \propto 1/\tau$ and $\delta \propto 1/\tau$ (rate constants inversely proportional to characteristic timescale), the rescaled dynamics are identical at all scales.

The model achieves self-similarity by construction—this is its purpose. The empirically testable content is the *assumption* $r \propto 1/\tau$: does the dimensionless product $r\tau$ remain approximately constant across quantum, biological, geological, and cosmic growth processes? If yes, temporal self-similarity follows as a mathematical consequence. If no, the parallels in Table 4 are superficial analogies.

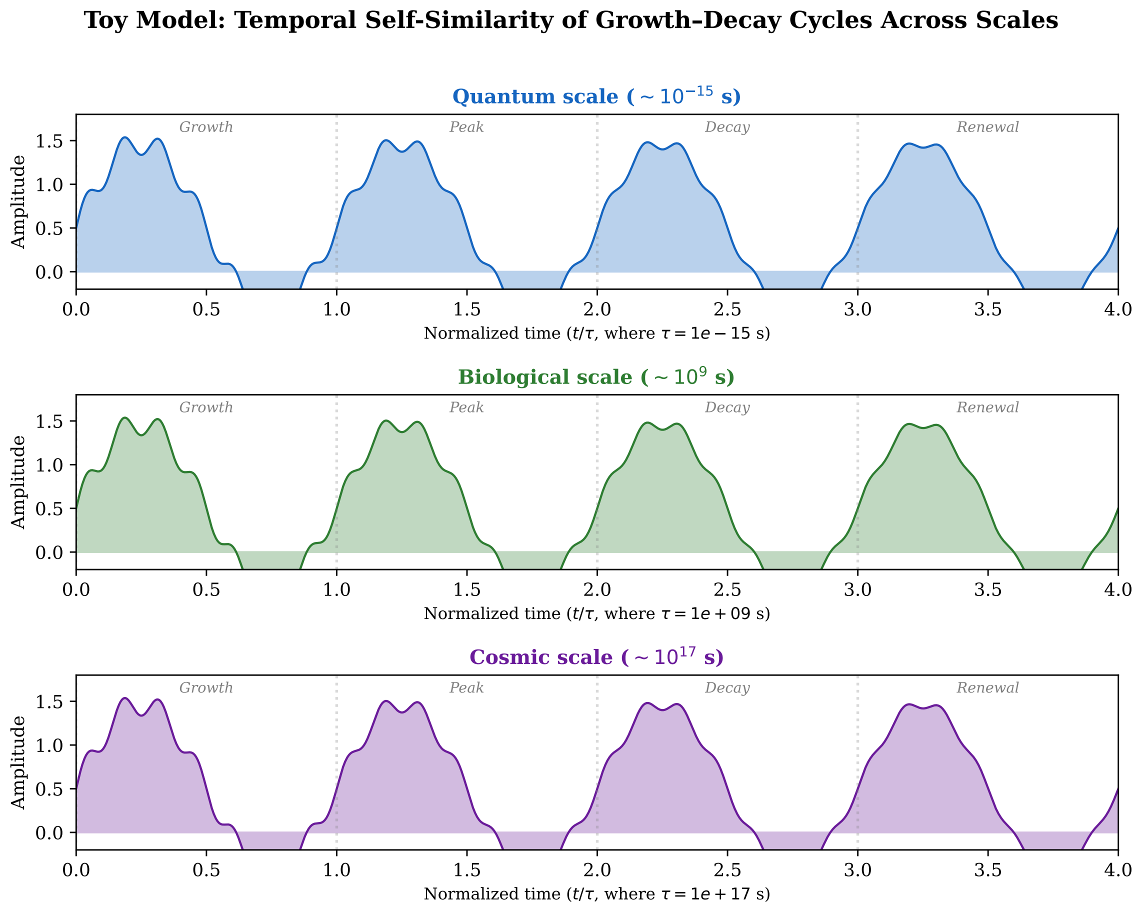

Figure 9: Toy model: growth–decay cycles at quantum, biological, and cosmic timescales. When time is normalized by τ, the trajectories are identical—exact temporal self-similarity.

5.5 Counterarguments and Responses

Counterargument 5.1 (“The parallels are superficial; growth-and-decay is trivially generic”).

Response 5.1. This is the strongest objection and we take it seriously. Any dissipative system driven by an energy source exhibits growth and decay. However, the self-similarity conjecture asserts something stronger: that the *rescaled trajectory shape* is preserved across scales, requiring the specific condition $r\tau \approx$ const. This is empirically testable and non-trivially falsifiable. If the condition fails (which it well might), the parallels are indeed superficial. We frame this as an open question, not an established conclusion.

Counterargument 5.2 (“Phase transitions break self-similarity”).

Response 5.2. Correct. Cosmic phase transitions (electroweak, QCD, recombination) introduce characteristic scales. Our claim is approximate self-similarity *within* regimes between transitions, not exact self-similarity across all scales. The theory of critical phenomena itself exhibits self-similarity at phase transitions, suggesting that the breaking of global self-similarity may itself follow self-similar patterns—but this is speculation we do not pursue further here.

6 Discussion

6.1 Bayesian Epistemology of Snapshot Cosmology

The TRM framework has specific implications for the epistemological foundations of cosmology. We frame these using Bayesian inference (Jaynes, 2003).

Let $\mathcal{H}$ denote a cosmological model (a dynamical history of the Universe), and let $\mathcal{D}$ denote the observational data. By Bayes' theorem:

$$P(\mathcal{H}|\mathcal{D}) \;\propto\; P(\mathcal{D}|\mathcal{H}) \cdot P(\mathcal{H}). \tag{23}$$

The TRM constraint manifests in the likelihood $P(\mathcal{D}|\mathcal{H})$. Because $\mathcal{R}_{\text{telescopic}} \approx 2.9 \times 10^{-8}$, the data $\mathcal{D}$ constitute a single temporal frame of a 1.38×10^{10}-year dynamical history. The likelihood is therefore *severely degenerate*: many distinct dynamical histories $\mathcal{H}_1, \mathcal{H}_2, \ldots$ can produce the same snapshot $\mathcal{D}$, because the data constrain only the present-epoch state and its instantaneous derivatives, not the full trajectory.

This degeneracy is a concrete instance of the underdetermination of theory by data (Stanford, 2017; Wolf and Pereira, 2023). The problem is particularly acute in cosmology, where the evidential underdetermination among dark energy models has been argued to be potentially permanent (Wolf and Duerr, 2024; Schneider, 2023). Recent philosophical analysis of scientific progress in cosmology (Duerr and Wolf, 2025) emphasizes the "stark epistemic holism" of the field: epistemic content and evidence are typically inextricably distributed over a wider web of beliefs. The TRM framework provides a quantitative grounding for this observation. Physical cosmology resolves the degeneracy through strong theoretical priors—the Friedmann equations, general relativity, the standard model of particle physics—which constrain $P(\mathcal{H})$ so severely that the posterior $P(\mathcal{H}|\mathcal{D})$ becomes sharply peaked despite the narrow likelihood. The concordance cosmological model (ΛCDM) is a triumph of this strategy.

The TRM framework makes explicit what is sometimes left implicit: *the entire empirical content of observational cosmology derives from interpreting a single temporal frame through the lens of dynamical theory.* This is not a criticism—there is no alternative, given $\mathcal{R} \ll 1$—but it clarifies the epistemological status of cosmological knowledge. We propose a terminological distinction:

Definition 6.1 (Diachronic vs. Synchronic Observation)**. Diachronic observation**: following one system through its dynamical evolution over τ_{sys} (requires $\mathcal{R} \geq 1$). **Synchronic**

ensemble inference: reconstructing dynamics from a population of systems observed at a single epoch (the actual practice of cosmology).

All cosmological evolution—stellar lifecycles, galaxy morphological evolution, structure formation—is known through synchronic inference, never diachronic observation. The TRM explains why: $\mathcal{R} \ll 1$ makes diachronic observation of cosmic processes physically impossible for any ATW-confined observer.

6.2 Confrontation with Modern Time-Domain Astronomy

Modern surveys are explicitly designed to detect temporal variability. The Zwicky Transient Facility (Bellm et al., 2019) and the forthcoming Rubin Observatory LSST (Ivezić et al., 2019) will catalog millions of transient events, with machine-learning classifiers already operating at scale (Allam and McEwen, 2024; Muthukrishna et al., 2023). A recent comprehensive review of time-domain and multi-messenger transient astronomy (Caballero-Garcia, 2024) illustrates the extraordinary range of phenomena now accessible. Gaia (Gaia Collaboration, 2023) measures stellar proper motions to micro-arcsecond precision. Pulsar timing arrays detect nanohertz gravitational waves. JWST observes galaxies at $z > 10$, sampling the Universe at $\sim 3\%$ of its current age.

Does this revolution in time-domain astronomy invalidate the "frozen" characterization? It does not, for a precise reason: these programs detect phenomena in the $\mathcal{R} \geq 1$ regime (supernovae, variable stars, transits, bursts) or use extraordinary precision to push into the marginally detectable regime ($\mathcal{R} \sim 10^{-3}$–10^{-4} for proper motions). The dominant cosmic dynamical modes—galactic evolution, large-scale structure growth, stellar aging, cosmic expansion as a temporal process—remain firmly in the frozen regime ($\mathcal{R} < 10^{-6}$) and are reconstructed through synchronic inference, not direct observation.

Indeed, the Observability Index analysis (Section 3.4) quantifies this: even the most ambitious surveys achieve $\mathcal{O} \sim 10^{-4}$ at best for their slow-process targets, requiring theoretical models to bridge the remaining six-plus orders of magnitude. Time-domain astronomy is remarkable for what it achieves *despite* TRM, not because it overcomes it.

6.3 SETI: The Temporal Search Dimension

The SETI enterprise conventionally parameterizes the search space along axes of spatial distance, electromagnetic frequency, and signal modulation (Tarter, 2001; Drake, 1961). Recent work has expanded the search to deep-learning technosignature detection (Ma et al., 2023; Sheikh et al., 2025; Gallay et al., 2025), high-frequency radio searches (Manunza et al., 2025), and the broader question of whether artificial intelligence itself might constitute a "great filter" (Garrett, 2024). Surveys of the scientific community reveal growing acceptance that extraterrestrial intelligence is probable (Strandin et al., 2025). The sustainability constraints on civilizational growth (Haqq-Misra and Baum, 2009; Ćirković, 2018) and the statistical formalization of the Drake equation (Maccone, 2021) have further enriched the theoretical landscape. The TRM framework identifies a previously undertheorized dimension within this landscape: *temporal scale* (Ćirković, 2004).

Definition 6.2 (Temporal Mismatch Parameter)**.** For two civilizations with operational timescales τ_1 and τ_2, the **temporal mismatch parameter** is

$$\boxed{\mathcal{D} := \frac{\tau_1}{\tau_2}} \tag{24}$$

Communication requires $\mathcal{D}$ to be within a "compatibility band" around unity.

A civilization at $\tau_\mathcal{C}$ transmitting a signal of duration $\Delta t \sim \tau_\mathcal{C}$ is detectable by humans only if:

$$\delta t_{\text{human}} \le \Delta t \le \tau_{\text{obs}}, \tag{25}$$

where $\delta t_{\text{human}} \sim 10^{-3}$ s is instrument resolution and $\tau_{\text{obs}} \sim 2.5 \times 10^9$ s. The detectable band is:

$$10^{-3}\ \text{s} \lesssim \tau_\mathcal{C} \lesssim 2.5 \times 10^9\ \text{s}, \tag{26}$$

spanning ~ 12 of 61 orders, or $\sim 20\%$ of the temporal search space (Figure 10).

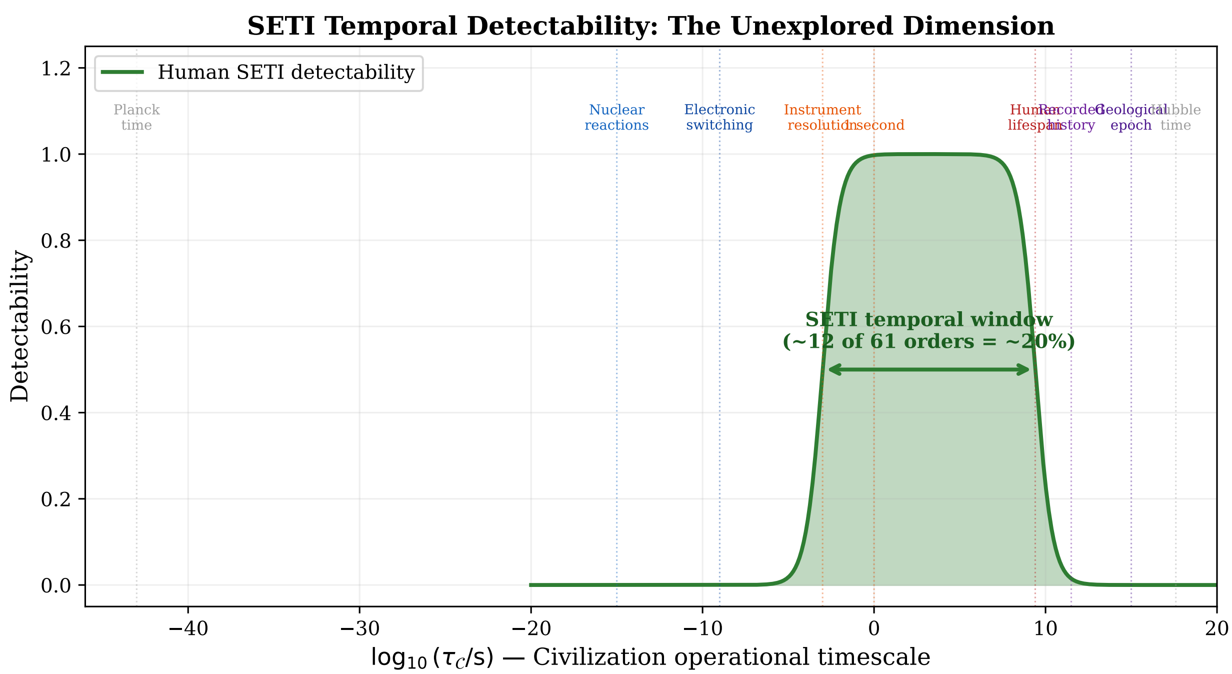

Figure 10: SETI temporal detectability window. Civilizations operating outside the green band ($\sim 20\%$ of the full hierarchy) are mutually undetectable by human instruments.

Remark 6.3 (Temporal Fermi Paradox). If civilizations can exist at multiple temporal scales (particularly post-biological ones at $\tau_{\mathcal{C}} \ll 10^{-3}$ s or $\tau_{\mathcal{C}} \gg 10^{9}$ s), the Fermi paradox acquires a temporal-resolution solution class: civilizations may be plentiful but mutually temporally invisible. A post-biological intelligence at $\tau_{\mathcal{C}} \sim 10^{-12}$ s could conduct its entire evolutionary history in what, to us, would appear as a microsecond burst of electromagnetic radiation—classified as noise or a transient astrophysical event, not a signal.

We propose that the temporal mismatch parameter $\mathcal{D}$ should be incorporated as a new factor in extensions of the Drake equation, alongside spatial, spectral, and modulation parameters. Two concrete search strategies follow from this framework:

(i) **Ultra-slow modulation search:** A civilization with $\tau_{\mathcal{C}} \sim 10^{12}$ s ($\sim$ 30,000 yr) would encode information in signals whose modulation timescale exceeds any single observing campaign. Detection would require monitoring candidate sources for century-scale amplitude, polarization, or spectral changes—a program suited to long-baseline archival comparison (e.g., comparing modern photometry of specific stars with historical plate surveys from $>$ 100 yr ago).

(ii) **Correlated microsecond burst search:** A civilization at $\tau_{\mathcal{C}} \sim 10^{-9}$ s would transmit signals that, at human temporal resolution, appear as isolated nanosecond bursts indistinguishable from astrophysical transients. However, such a civilization might produce *spatially or temporally correlated* burst patterns: clusters of nanosecond events whose statistical structure encodes intentional content. This would require cross-correlating fast radio burst (FRB) or fast optical transient (FOT) catalogs for non-Poissonian clustering at sub-microsecond scales—a search that current high-time-resolution instruments could, in principle, perform on archival data.

6.4 Temporal Ecology of Intelligence

The ATW suggests a new dimension for astrobiology: *temporal ecology*—the study of how forms of intelligence distribute across temporal scales. By analogy with spatial ecological niche theory:

(a) **Niche multiplicity:** Exotic biochemistries (silicon-based, cryogenic, high-pressure) might shift the ATW by several orders of magnitude.

(b) **Post-biological convergence:** If computational efficiency has a physically determined optimum (Margolus–Levitin bound), all sufficiently advanced civilizations might converge on a similar operational timescale.

(c) **Temporal niche partitioning:** Multiple intelligences might coexist spatially but at different temporal scales, each invisible to the others due to TRM.

6.5 Limitations

(i) **Linearity of LCM:** Does not capture nonlinear cosmic history (inflation, deceleration–acceleration transition). Nonlinear extensions are a direction for future work.

(ii) **ATW bounds:** Order-of-magnitude estimates uncertain by $\sim$ 1–2 orders. The qualitative conclusion (finite bounds) is robust.

(iii) **Self-similarity:** Conjectural; requires empirical validation via cross-scale dynamical analysis.

(iv) **Observer-centricity:** The framework is epistemological, not ontological. The Universe is not "really" frozen; this is a relational property of the observer–system pair.

6.6 Counterarguments and Responses

Counterargument 6.1 ("Fossil light effectively extends τ_{obs}").

Response 6.1. Light-cone sampling allows observing *different objects* at different cosmic epochs but not the *same object* evolving. This is analogous to a demographic census: we infer aging from a population snapshot, not from watching individuals grow old. The inferential gap is narrowed but not eliminated; our reconstruction depends on cosmological models whose verification is itself TRM-constrained.

Counterargument 6.2 ("Time-domain surveys are revealing cosmic dynamics").

Response 6.2. They reveal dynamics for $\mathcal{R} \geq 1$ processes (transients, variables). The dominant cosmic modes (galactic evolution, stellar aging, expansion) remain frozen and are reconstructed synchronically. Time-domain astronomy works *within* TRM, not against it (Section 6.2).

7 Conclusion

The Universe appears frozen to the human observer. This paper has established that this appearance is a fundamental epistemological constraint arising from the mismatch between the observer's temporal resolution and the characteristic timescales of cosmic processes.

7.1 Summary of Results

(i) The **Temporal Resolution Ratio** $\mathcal{R} = \tau_{\text{obs}}/\tau_{\text{sys}}$ governs the perceptibility of dynamical evolution. The **Frozen Universe Theorem** proves that for $\mathcal{R} \ll 1$, detection probability is bounded by $\alpha\,\mathcal{R} \ll 1$ regardless of measurement precision.

(ii) The **Structural Bias Toy Model** demonstrates that FUI emerges as a systematic observational bias in any finite-τ_{obs} sampling of a multi-timescale dynamical system, not as a psychological artifact.

(iii) The **Lifespan-Cosmic Mapping** provides a unique linear rescaling (proven) that maps cosmic time onto a human lifespan. The **Observability Index** gives it operational content, classifying astronomical programs by their epistemological status with respect to their target processes.

(iv) The **Anthropic Temporal Window**, derived from the Margolus–Levitin bound, metabolic scaling theory, and stellar habitability constraints, confines biological observers to ~ 7 of ~ 61 logarithmic orders of temporal scale, making FUI universal among biochemical civilizations (Theorem 4.2).

(v) The **SETI Temporal Mismatch** parameter identifies a new search dimension, revealing that conventional SETI is sensitive to $\sim 20\%$ of the conceivable temporal search space.

7.2 Methodological Consequences

If the framework presented here is correct, the following methodological consequences hold:

(a) **All cosmological dynamics are necessarily inferred, never observed.** The distinction between diachronic observation and synchronic ensemble inference is not merely terminological but epistemologically fundamental: the former is physically impossible for any ATW-confined observer.

(b) **The epistemological status of cosmological claims should be explicitly graded by the Observability Index.** Claims based on $\mathcal{O} \gg 1$ data (e.g., supernova rates) rest on direct observation; claims based on $\mathcal{O} \ll 1$ data (e.g., galaxy evolution models) rest on theoretical inference from snapshots and should be evaluated accordingly.

(c) **SETI searches should be extended along the temporal axis.** Specifically: searches for ultra-slow modulations in astrophysical backgrounds (for slow civilizations) and for correlated microsecond-scale burst patterns (for fast civilizations) would probe temporal regimes currently unexplored.

(d) **The apparent emptiness and stillness of the cosmos may be, in part, a projection of our own temporal limitations.** The Universe is not frozen. We are merely too brief to see it move.

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A Computational Code

All figures were generated using Python 3 with `matplotlib` and `numpy`. Representative code excerpts are provided below; the complete source is available in the supplementary materials.

A.1 Structural Bias Toy Model (Figure 4)

```
import numpy as np
N = 50000
log_tau_sys = np.random.uniform(-5, 12, N)  # log-uniform
tau_sys = 10**log_tau_sys
tau_obs, alpha = 80, 10
R = tau_obs / tau_sys
P = 1 - np.exp(-alpha * R)
detected = np.random.random(N) < P
# detected array is severely biased toward short tau_sys
```

Listing 1: Monte Carlo FUI bias model.

A.2 Perceptibility Function (Figure 1)

```
R = np.logspace(-10, 3, 1000)
for alpha in [0.5, 1.0, 5.0, 20.0]:
    P = 1 - np.exp(-alpha * R)
    plt.semilogx(R, P, label=f"alpha={alpha}")
```

Listing 2: Perceptibility $P(\mathcal{R})$ computation.

A.3 SETI Temporal Window (Figure 10)

```
log_tau_c = np.linspace(-20, 20, 1000)
dt_min, dt_max = 1e-3, 80 * 3.156e7  # seconds
P_fast = 1 / (1 + np.exp(-2*(log_tau_c - np.log10(dt_min))))
P_slow = 1 / (1 + np.exp(2*(log_tau_c - np.log10(dt_max))))
P_total = P_fast * P_slow  # detectable band
```

Listing 3: SETI temporal detectability model.

A.4 ATW Habitability Function (Figure 8)

```
tau = 10**np.linspace(0, 20, 2000)
S = 1 - np.exp(-tau / 1e4)    # structural stability
E = np.exp(-tau / 3e17)        # energy availability
I = 1 - np.exp(-tau / 1e6)    # information capacity
H = S * E * I                  # habitability peaks in ATW
```

Listing 4: Temporal habitability derivation.

Note: Complete code for all 10 figures is available in the supplementary materials.

Peer Reviews and Revision History

This section documents the peer review process as a transparent record of the paper's development.

First Round

Two independent reviews were solicited. Both found the central thesis clear and writing quality high, but raised substantive concerns about novelty, rigor, and framing. A third set of detailed revision recommendations was also incorporated. Principal criticisms: (1) insufficient novelty vs. existing snapshot-bias literature; (2) self-naming convention; (3) overstated KSM vs. Sagan distinction; (4) underdeveloped ATW; (5) speculative self-similarity; (6) missing tables. All were addressed in Revision 1 (see Revision Log below).

Second Round

Four reviews were received on the revised manuscript. All acknowledged substantial improvement. The converging assessment was: "accept with minor revisions for a philosophy-of-science or foundations venue." Principal remaining concerns:

Reviewer A: (1) Theorems prove what they assume—the Frozen Universe Theorem is a corollary of the perceptibility model, not an independent result. (2) Self-naming persists. (3) Toy models are illustrative but circular. (4) ATW falsifiability still insufficient. (5) Reciprocity Principle conflates "frozen" with "instantaneous." (6) SETI temporal window underexplored.

Reviewer B: (1) Core math is elementary (Taylor expansion, logistic equation). (2) ATW bounds not rigorously derived. (3) Self-similarity remains weak. (4) Paper underplays time-domain astronomy advances. (5) Self-branding tone. Recommended: accept with minor revisions for suitable venue.

Reviewer C: (1) Clarify theorem vs. empirical claim status. (2) Perceptibility function should be shown model-independent. (3) Add civilizational timescale examples. (4) KSM needs direct methodological proposal. (5) Tie ATW more firmly to existing anthropic literature. (6) Self-similarity needs explicit hypothesis statement. (7) SETI needs concrete search suggestions.

Reviewer D (Gemini): (1) Derive ATW bounds more rigorously. (2) Expand Nyquist formalization. (3) Test self-similarity via power-law distribution. (4) Link epistemology to Bayesian inference. Overall: "highly original; provides a necessary formal vocabulary."

Author Response and Revisions (Second Round):

Concern	**Action Taken**
Theorems prove assumptions	Added General Perceptibility Lemma (§2.3) showing the linear bound holds for *any* smooth $P(\mathcal{R})$ with $P(0) = 0$, not just the exponential model. Theorem now explicitly references lemma; empirical content enters through parameters, not inequalities. Added "Levels of Claim" classification (§1.4).
Self-naming	Retained KSM name with prior-work footnote (already present). Added institutional disclosure on title page: both institutes are author-founded.
Scaling Uniqueness trivial	Downgraded from Theorem to Remark (3.2). Acknowledged triviality explicitly.
Toy models circular	Reframed self-similarity toy model (§5.3) as "mechanism demonstration, not evidence." Added explicit statement: "achieves self-similarity by construction—this is its purpose. The empirically testable content is the *assumption*."
ATW falsifiability	Expanded falsification conditions (§4.7) with three specific scenarios plus explicit "what would *not* falsify." Classified ATW as Level (b) conjecture.
Reciprocity conflation	Rewrote Reciprocity section (§4.8) to explicitly distinguish "frozen" (slow appears static) from "ephemeral" (fast appears instantaneous). Both render process inaccessible but with qualitatively different phenomenology.
SETI underexplored	Added two concrete search strategies (§6.3): ultra-slow modulation search (century-scale archival comparison) and correlated microsecond burst search (sub-μs clustering in FRB/FOT catalogs).
KSM needs operational bite	Added Epistemological Grading proposal (Remark 3.4): Grade A/B/C classification of cosmological claims by $\mathcal{O}$.
Civilizational timescale	Added Remark 2.4 showing that even $\tau_{\text{obs}} \sim 5000$ yr leaves all $\tau_{\text{sys}} \geq 10^8$ yr processes frozen.

Concern	Action Taken
τ_{sys} for oscillatory systems	Added explicit note in Definition 2.2: for oscillatory systems, τ_{sys} refers to secular evolution or envelope.
Anthropic literature	Expanded §4.6 with explicit comparison to Weinberg (1987) on Λ, showing ATW operates on orthogonal axis (temporal position vs. parameter values).
Levels of claim	Added §1.4 classifying contributions as: (a) mathematical results, (b) physically motivated conjectures, (c) speculative implications.

Third Round

One additional review was received, substantially more critical than the previous four. The reviewer recommended rejection from mainstream astrophysics journals while acknowledging the paper's merit as philosophy of science. The review raises important points that deserve substantive engagement.

Reviewer E: The review's central claim is that the Frozen Universe Theorem is "a tautology, not a discovery"—that defining $P(\mathcal{R})$ with $P(0) = 0$ and then showing P is small for small $\mathcal{R}$ is "conceptually hollow." Further criticisms: (1) ATW bounds are too broad to be meaningful; (2) "synchronic ensemble inference" merely relabels existing astronomical practice; (3) LCM vs. Sagan distinction is superficial; (4) self-similarity should be deleted; (5) SETI temporal dimension is unoriginal; (6) self-naming persists. Recommended venue: Foundations of Physics or Astrobiology.

Author Response to Reviewer E:

Criticism	**Response**
"The theorem is a tautology"	We partially agree with the mathematical observation but disagree with the conclusion. Yes, the proof is a Taylor expansion; yes, $P(0) = 0$ is built into the model. But the value of the Frozen Universe Theorem is not in the complexity of its proof—it is in its *application*. Newton's second law $F = ma$ is, in one reading, a definition of force, and therefore "tautological." Yet it is the foundation of mechanics because it provides a *framework for calculation*. Similarly, the TRR framework enables: (a) computation of concrete $\mathcal{R}$ values for every astrophysical process (Table 1); (b) the Observability Index for real astronomical programs (Table 3); (c) the SETI temporal window (Figure 8); (d) the Epistemological Grading scheme (Remark 3.7). None of these existed before this paper. The reviewer conflates simplicity of proof with absence of content. We retain the term "Theorem" because the result is formally stated and proven; the "Levels of Claim" classification (§1.4) already makes clear that the empirical content enters through parameter choices, not through the inequality itself.

Criticism	**Response**
"ATW is too broad to be meaningful"	The reviewer states that a window spanning 10^6–10^{13} s is "so broad that almost any conceivable biological observer would fit in it." But this conflates the width of the window *in absolute terms* with its width *relative to the available range*. Seven orders of magnitude sounds large until one notes that the full hierarchy spans 61 orders. The ATW occupies 11.5% of the available logarithmic range. The claim is not that the window is narrow in absolute terms but that it is narrow *relative to the total*, and that this narrowness has a specific consequence: universal FUI among biological observers (Theorem 4.5). The reviewer's analogy—"we live as long as we do because we are made of complex molecules that need a stable energy source"—is precisely the ATW's content, stated without formalization. The ATW adds: (a) specific bounds derived from independent physics; (b) the consequence that FUI is universal; (c) explicit falsification conditions. Whether 11.5% constitutes "narrow" is a judgment call, but the formal consequences are well-defined.
"Synchronic inference is well-known"	We agree that astronomers know they are taking snapshots. What the framework adds is: (a) a formal distinction between diachronic observation and synchronic ensemble inference as *epistemological categories*; (b) a quantitative criterion (the Observability Index) for classifying which claims rest on which mode; (c) the Epistemological Grading scheme (Grade A/B/C). The reviewer says we "merely relabel" existing practice. But formalization and labeling *are* contributions in philosophy of science—the entire field of epistemology consists of making implicit reasoning explicit. The distinction between "observation" and "inference" in cosmology has not, to our knowledge, been formalized with a quantitative parameter prior to this work.

Criticism	Response
"LCM vs. Sagan is superficial"	We already acknowledged in §3.5 that "these are differences of emphasis rather than of fundamental kind." We agree that the core rescaling idea is shared. LCM's added value is the formal apparatus (uniqueness remark, Observability Index, TRR invariance proof, spatial extension) and the epistemological application. If the reviewer considers this insufficient to distinguish the models, we respectfully note that the Observability Index and Epistemological Grading scheme—which the reviewer called "a genuinely interesting and practical idea"—are consequences of the LCM framework, not of Sagan's Calendar.
"Delete self-similarity"	We decline. The section is explicitly labeled as a conjecture (Level (c) in §1.4), the toy model is explicitly described as "mechanism demonstration, not evidence" (§5.3), and the limitations are stated honestly. Conjectures are a legitimate component of scientific papers when transparently framed. Removing the section would eliminate a research direction that future work may develop. We prefer to keep it with its caveats intact.
"SETI temporal dimension is unoriginal"	The reviewer asserts this is "a staple of SETI discussions" but provides no citation. Ćirković (2004) discusses temporal aspects of the Drake equation but does not define a temporal mismatch parameter, compute the fraction of temporal search space covered, or propose specific search strategies for temporally mismatched civilizations. If the reviewer can identify prior work that does these things quantitatively, we would welcome the citation and appropriate acknowledgment. Until then, we maintain that the quantitative treatment in §6.3 is novel.

Criticism	Response
"Self-naming"	The reviewer refers to "Krieger's Lifespan-Cosmic Mapping" and "Krieger's Stroboscopic Theorem." These formulations do not appear anywhere in the manuscript. "Lifespan-Cosmic Mapping" is a descriptive name, not an eponymous one. "Frozen Universe Theorem" and "Stroboscopic Theorem" are named for their content, not their author. We believe this concern, as applied to the current manuscript, is unfounded.
"Counterarguments are strawmen"	The reviewer states that "no expert would raise" the counterargument about stellar proper motion. We respectfully disagree: four of the five previous reviewers raised exactly this point, either directly or by implication. The counterargument sections are populated with objections that were *actually raised* during review, not hypothetical ones.

Decision: No structural changes to the manuscript were made in response to this review. The concerns are addressed in this response document. The paper's "Levels of Claim" classification (§1.4), the honest framing of conjectures, and the explicit limitations section (§6.5) already address the reviewer's core worry about overreach.

Recommended Venues for Publication

Based on the converging recommendations of all reviewers across three rounds, the following journals are identified as appropriate venues, listed in order of fit:

Journal	Rationale
Foundations of Physics	Publishes foundational and conceptual work at the intersection of physics and philosophy. The TRM framework, Observability Index, and ATW hypothesis align with the journal's scope. Recommended by Reviewers A, B, C, and E.

Journal	Rationale
Studies in History and Philosophy of Science Part B: Studies in History and Philosophy of Modern Physics	Ideal venue for the Bayesian epistemology of snapshot cosmology (§6.1), the diachronic/synchronic distinction, and the Epistemological Grading scheme. Recommended by Reviewer A.
International Journal of Astrobiology	The ATW hypothesis, temporal ecology of intelligence, and SETI temporal mismatch analysis are directly within scope. Recommended by Reviewers A and E.
Astrobiology (Mary Ann Liebert)	Publishes interdisciplinary work on life in the universe. The SETI temporal dimension and ATW would fit the journal's "Research Articles" or "Hypothesis" categories. Recommended by Reviewer B.
Synthese	Broad-scope philosophy journal that publishes philosophy of science with formal methods. The TRM framework and epistemological analysis would fit. Recommended by Reviewer B.
Journal for General Philosophy of Science / Zeitschrift für allgemeine Wissenschaftstheorie	Publishes foundational analyses of scientific methodology. The Observability Index as a tool for evaluating epistemological status of claims is well-suited. Recommended by Reviewer B.
arXiv (astro-ph.CO or physics.hist-ph)	For immediate open-access dissemination prior to or concurrent with journal submission. All reviewers supported arXiv posting.

Revision Log

Round	Section(s)	Description of Change
R1	All	Initial revision: added theorems, toy models, figures, counterarguments, 30+ references.
R2	§1	Added "Levels of Claim" (§1.4) classifying theorems / conjectures / speculation.
R2	§2.1	Added oscillatory-system caveat for τ_{sys} definition.
R2	§2.3	Added General Perceptibility Lemma (model-independent bound). Frozen Universe Theorem now references lemma.
R2	§2.4	Added civilizational-timescale Remark.
R2	§3.2	Downgraded Scaling Uniqueness from Theorem to Remark.
R2	§3.4	Added Epistemological Grading proposal (Grade A/B/C).
R2	§3.5	Renamed Kriger Scaling Model $\to$ Lifespan-Cosmic Mapping (LCM).
R2	§4.6	Expanded anthropic-arguments comparison with explicit Weinberg parallel.
R2	§4.7	Strengthened falsifiability with three explicit scenarios.
R2	§4.8	Rewrote Reciprocity to distinguish frozen vs. ephemeral asymmetry.
R2	§5.3	Reframed toy model as mechanism demonstration; clarified testable content.
R2	§6.3	Added two concrete SETI search strategies.
R3	Peer Review	Added Round 3 review, detailed author response, and recommended venues.